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also launched a couple of Power8 Linux servers, whose unique selling proposition is their ability to accelerate AI, deep learning and advanced analytics applications. The servers apparently move data five times faster than competing platforms using NVIDIA's NVLink high-speed interconnect technology.

IBM is also trying to do something more about reducing the amount of computing power and time that deep learning requires. Their Watson Research Center believes that it can reduce these using theoretical chips called resistive processing units or RPUs that combine a central processing unit (CPU) and non-volatile memory. The team claims that such chips can accelerate data speeds exponentially, resulting in systems that can do tasks like natural speech recognition and translation between all world languages.

Currently, neural networks like DeepMind and Watson need to perform billions of tasks in parallel, requiring numerous CPU memory calls. Placing large amounts of resistive random access memory directly onto a CPU would solve this, as such chips can fetch data as quickly as these can process it, thereby reducing neural network training times and power required.

The research paper claims that, "This massively-parallel RPU architecture can achieve acceleration factors of 30,000 compared to state-of-the-art microprocessors—problems that currently require days of training on a data centre size cluster with thousands of machines can be addressed within hours on a single RPU accelerator." Although these chips are still in the research phase, scientists say that these can be built using regular complementary metal-oxide semiconductor technology.



Big Sur is designed around operational efficiency and serviceability (Image courtesy: <https://code.facebook.com>)

Locally-done deep learning.

Speaking of reducing processing power and learning times, can you imagine deep learning being performed locally on a mobile phone, without depending on the Cloud? Well, Apple revealed at 2016's Worldwide Developer's Conference (WWDC) that it can do precisely that. The company announced that it is applying advanced, deep learning techniques to bring facial recognition to iPhone, and it is all done locally on the device. Some of this success can be attributed to Perceptio, a company that Apple acquired last year. Perceptio is developing deep learning tech that allows smartphones to identify images without relying on external data libraries.

Facebook intends to understand your intent

Facebook Artificial Intelligence Research (FAIR) group has come out with innumerable innovations, which are so deeply woven with their products that we do not even realise that we are using deep learning every time we use Facebook. FBLearn Flow is Facebook's internal platform for machine learning. It combines several machine learning models to process data points drawn from the activity of the site's users, and makes predictions such as which user is in a photograph, which post is spam and so on. Algorithms that come out of FBLearn Flow help Facebook to identify faces in photos, select content for your news feed and more.